Final Report

Internship Project

Project Title: Customer Churn Analysis and Prediction

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Project Period: March 01, 2026 – April 01, 2026

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Introduction

Customer retention has become a major priority for organizations operating in highly competitive industries such as telecommunications. Acquiring new customers is often more costly than retaining existing ones, making churn reduction a key strategic objective. With the availability of large-scale customer data, organizations can leverage analytics to better understand customer behavior and proactively address churn risks.

This project focuses on analyzing a telecom customer dataset to explore behavioral patterns and identify key drivers of churn. Using exploratory data analysis and machine learning techniques, the study aims to uncover meaningful insights and build a predictive framework that can support strategic business decisions and improve customer retention outcomes.

1. Project Definition

1.1. Business Problem Statement

Customer retention is a major challenge in the telecommunications industry. Customer churn leads to revenue loss, increased customer acquisition costs, and reduced business growth. This project aims to analyze the key factors influencing customer churn and develop a predictive model to identify customers at risk of leaving. The insights from this study will support data-driven decision-making to reduce churn and improve customer retention strategies.

1.2. Project Objectives

The main objectives of this project are:

- To analyze customer behavior and identify churn patterns
- To determine the key drivers influencing customer churn
- To develop a predictive model to identify high-risk customers
- To provide actionable business recommendations to reduce churn

1.3. Key Business Questions

This project seeks to answer the following questions:

- What factors have the strongest impact on customer churn?
- Is there a relationship between customer service calls and churn?

- Does having an international plan increase churn risk?
- Which customer segments are more likely to churn?
- What strategies can reduce customer churn?

1.4. Success Metrics (KPIs)

The success of this project will be evaluated using:

- Customer churn rate
- Model accuracy
- Precision, recall, and F1-score
- Number of high-risk customers correctly identified
- Key feature importance insights

2. Dataset Description

The dataset used in this study is the Telecom Customer Churn dataset, which contains customer-level information from a telecommunications company. The data is divided into two parts:

- Training dataset (churn-bigml-80.csv): 2,666 observations
- Testing dataset (churn-bigml-20.csv): 667 observations

The dataset includes 20 variables representing customer demographics, service plans, usage behavior, and customer service interactions.

The target variable in this study is:

- **Churn (True/False):** Indicates whether a customer has left the service.

The independent variables can be grouped into the following categories:

- **Demographic and Account Information:** State, account length, area code
- **Service Plan Features:** International plan, voice mail plan, number of voice mail messages
- **Usage Behavior Variables:** Total day, evening, night, and international minutes, calls, and charges

- **Customer Interaction Variable:** Number of customer service calls

This dataset is suitable for both exploratory data analysis and predictive modeling to identify patterns and key drivers of customer churn.

3. Data Understanding

3.1. Data Structure Overview

The training dataset contains 2,666 observations and 20 variables. The dataset includes a mix of numerical and categorical variables representing customer demographics, service plans, usage behavior, and customer interactions. Initial inspection shows that most variables are numeric, while a few are categorical (e.g., state, international plan, voice mail plan). No major structural issues were observed during the initial review.

3.2. Missing Values Analysis

A review of missing values shows that the dataset does not contain any missing observations. Therefore, no imputation or data cleaning was required at this stage.

3.3. Duplicate Records

No duplicate records were found in the dataset, indicating that each observation represents a unique customer.

3.4. Target Variable Distribution

The target variable (Churn) is imbalanced. Most customers did not churn, while a smaller proportion left the service. This imbalance is typical in churn datasets and will be considered during the modeling phase to ensure accurate evaluation of predictive performance.

3.5. Descriptive Statistics Overview

Descriptive statistics indicate variability across customer usage patterns, including call minutes, number of calls, and service usage. These variations suggest potential differences in customer behavior, which will be further explored during the exploratory data analysis phase.

4. Data Cleaning & Preparation

4.1. Drop Unnecessary Columns

To prepare the dataset for analysis and modeling, initial preprocessing steps were performed. The “State” variable was removed because it does not provide meaningful predictive value and may introduce unnecessary noise into the model.

4.2. Convert Categorical Variables

Categorical variables such as international plan, voice mail plan, and churn were converted into binary numerical format to support statistical analysis and machine learning modeling.

4.3. Outlier Check

A preliminary review of outliers was conducted using boxplots. Although minor outliers were observed in some usage-related variables, they were retained to preserve realistic customer behavior patterns.

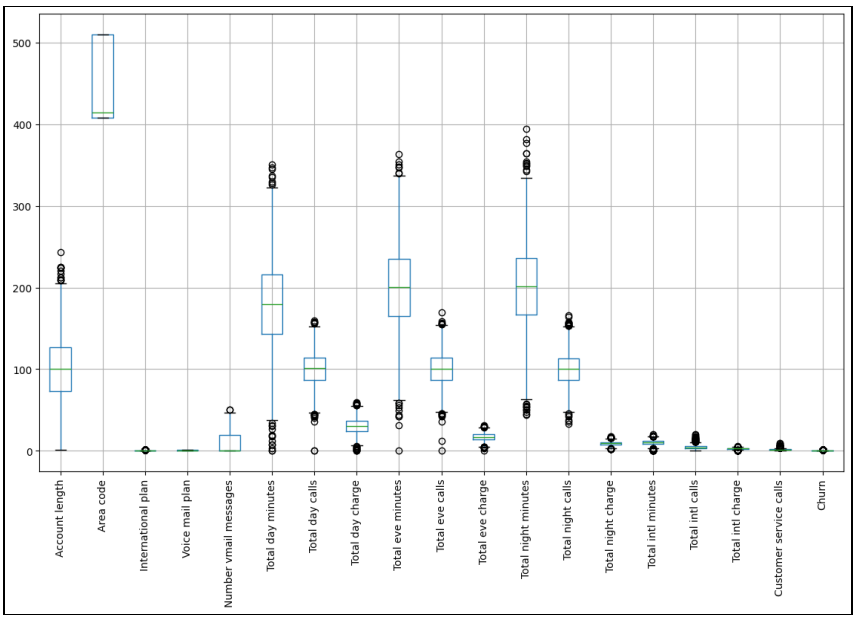

Figure 1. Outlier Assessment

A visual inspection using boxplots revealed the presence of outliers across several usage-related variables, including day, evening, night, and international call metrics. These outliers are expected in real-world telecom data, as customer usage naturally varies across individuals.

Additionally, some extreme values were observed in customer service call frequency, which may reflect dissatisfied customers and provide meaningful insights for churn analysis. Since

these values represent realistic customer behavior rather than data errors, no outliers were removed from the dataset.

After preprocessing, the dataset was confirmed to be clean and ready for exploratory data analysis and predictive modeling.

5. Exploratory Data Analysis (EDA)

5.1. Churn Distribution

Figure 2 illustrates that most customers did not churn, while a smaller proportion left the service. This indicates an imbalanced dataset, which is common in churn-related problems and highlights the importance of careful model evaluation in later stages.

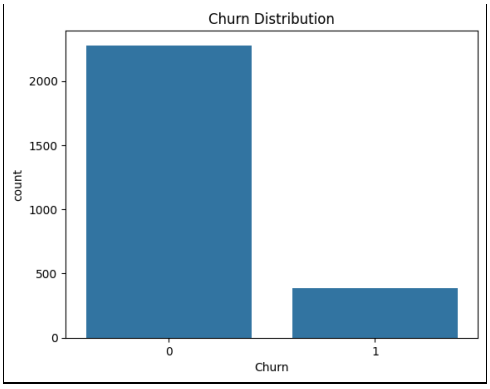

Figure 2. Churn Distribution

5.2. Churn vs Customer Service Calls

In figure 3 the boxplot analysis reveals a clear difference between churned and retained customers in terms of customer service interactions. Customers who churned had a higher median number of service calls and showed greater variability compared to non-churn customers.

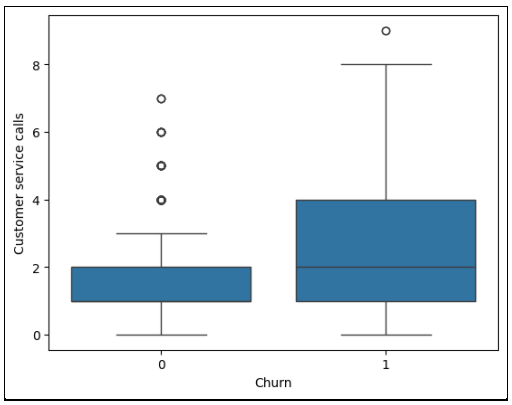

Figure 3. Churn vs Customer Service Calls

This pattern suggests that frequent customer service interactions may indicate dissatisfaction or unresolved service issues, making customer service calls a strong potential predictor of churn. From a business perspective, customers with high service call frequency could be considered an early warning segment for targeted retention strategies.

5.3. Churn vs International Plan

In figure 4 the analysis shows that customers with an international plan have a noticeably higher churn rate compared to those without the plan. Although the number of customers with international plans is smaller, a larger proportion of them tend to leave the service.

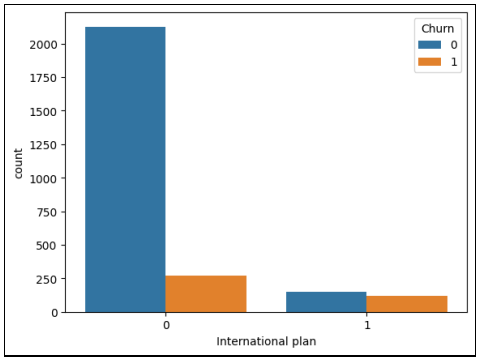

Figure 4. Churn vs International Plan

This finding suggests that customers using international services may be more sensitive to pricing or service quality issues. From a business perspective, customers with international plans represent a higher-risk segment and may benefit from targeted retention strategies such as customized pricing, loyalty incentives, or service improvements.

5.4. Churn vs Voice Mail Plan

In figure 5 the analysis indicates that customers with a voice mail plan are less likely to churn compared to those without the plan. Customers without voice mail services show a noticeably higher proportion of churn.

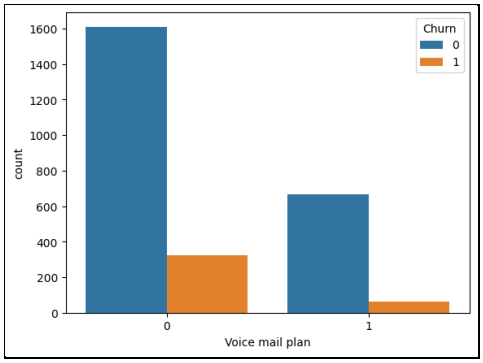

Figure 5. Churn vs Voice Mail Plan

This finding suggests that customers who subscribe to additional services may be more engaged and have stronger loyalty to the provider. From a business perspective, promoting value-added services such as voice mail plans may support customer retention and reduce churn risk.

5.5. Usage Behavior Analysis

In figures 6 – 9 the boxplot analysis reveals noticeable differences in usage behavior between churned and retained customers. Customers who churned generally showed higher usage levels across multiple categories, particularly in daytime call minutes.

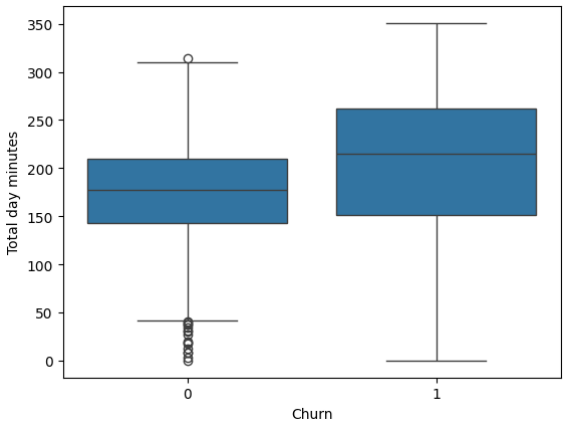

Figure 6. Churn vs total day minutes

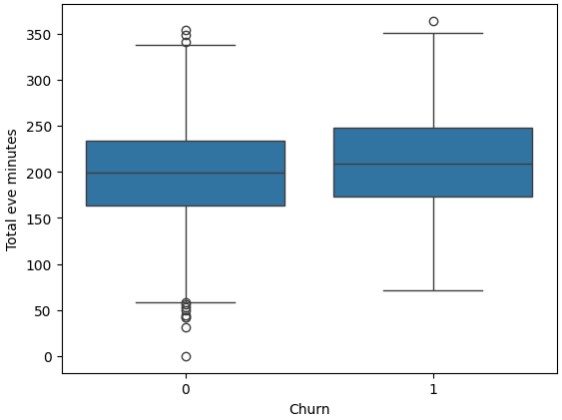

Figure 7. Churn vs total eve minutes

Daytime usage demonstrated the strongest separation between the two groups, suggesting that heavy daytime users may be more sensitive to pricing or perceived service value. Differences were also observed in evening, night, and international usage, although these effects were less pronounced.

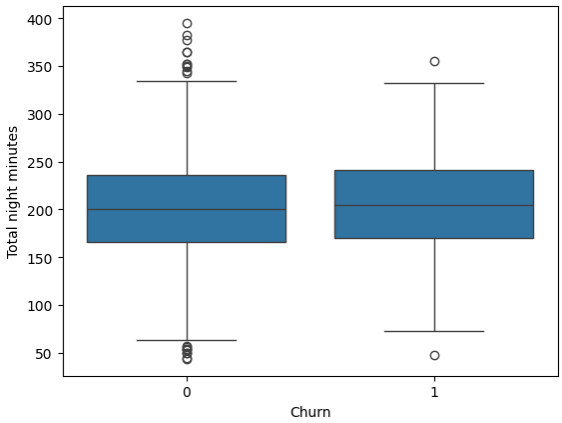

Figure 8. Churn vs total night minutes

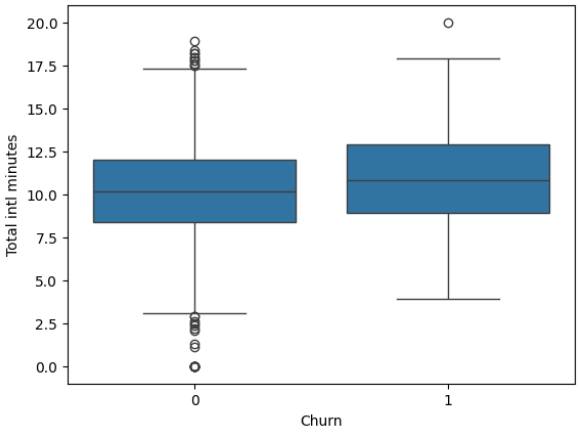

Figure 9. Churn vs total intl minutes

From a business perspective, high-usage customers represent an important segment for targeted retention strategies, including pricing optimization, personalized plans, and proactive engagement efforts.

5.6. Correlation Analysis

I figure 10 the correlation heatmap revealed strong relationships between call minutes and their corresponding charges across all time periods, which is expected since charges are directly calculated from usage levels. These highly correlated variables indicate some redundancy in the dataset but do not negatively impact tree-based modeling approaches.

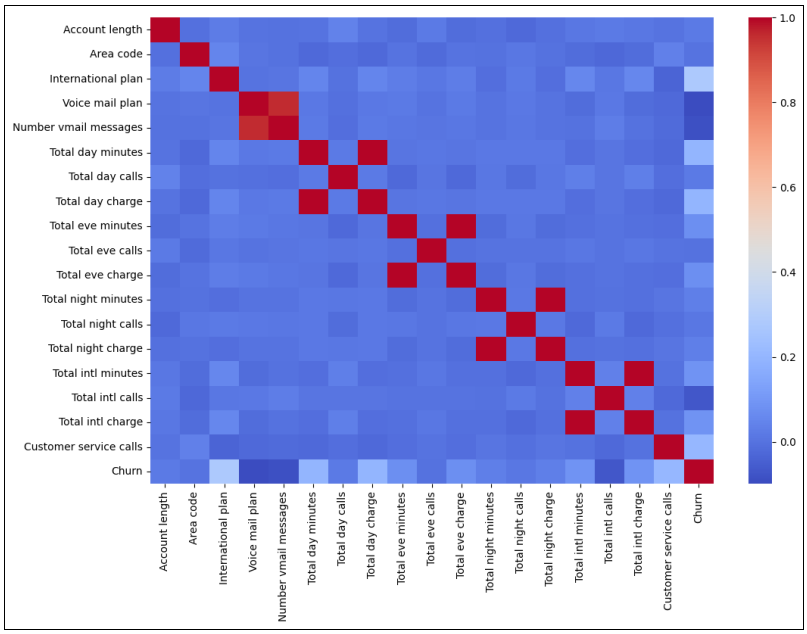

Figure 10. Correlation Analysis

In relation to churn, customer service calls showed the strongest positive correlation, followed by international plan and total daytime minutes. In contrast, voice mail plan demonstrated a negative relationship with churn, suggesting a protective effect. Overall, the results indicate that behavioral and service-related factors are more strongly associated with churn than demographic characteristics.

The exploratory data analysis (EDA) identified several key factors strongly associated with customer churn. Among all variables, customer service calls emerged as the most significant indicator, with churned customers consistently showing higher interaction frequency, suggesting dissatisfaction and unresolved service issues. Customers with international plans also demonstrated a higher likelihood of churning, potentially reflecting pricing sensitivity or

service-related concerns. Additionally, higher daytime call usage was observed among churned customers, indicating that heavy users may be more sensitive to perceived service value and pricing structures.

Conversely, customers with a voice mail plan were less likely to churn, suggesting that engagement with additional services may strengthen customer loyalty. Overall, the findings highlight behavioral and service-related factors as primary drivers of churn and provide a strong foundation for predictive modeling in the next phase.

6. Predictive Modeling (Machine Learning)

6.1. Baseline Model (Logistic Regression)

To predict customer churn, classification models were developed using the prepared dataset. A logistic regression model was first implemented as a baseline to evaluate initial predictive performance before applying more advanced techniques.

A logistic regression model was implemented as a baseline classifier to evaluate initial predictive performance. The model achieved an overall accuracy of approximately 85.6%, which appears reasonable at first glance. However, further evaluation revealed limitations in identifying churned customers.

The confusion matrix showed a high number of false negatives, indicating that many churned customers were incorrectly classified as non-churn. This is reflected in the low recall score (0.16) for churn prediction, suggesting that the model struggled to effectively detect customers at risk of leaving.

From a business perspective, recall is particularly important in churn prediction because failing to identify at-risk customers can result in significant revenue loss. Therefore, despite acceptable accuracy, logistic regression was not considered sufficient for this task and more advanced models were explored.

6.2. Main Model (Random Forest)

A Random Forest classifier was then applied as the primary model due to its strong performance in classification tasks and its ability to capture complex relationships between variables.

The Random Forest classifier demonstrated strong predictive performance and significantly outperformed the baseline logistic regression model. The model achieved an accuracy of approximately 95.5%, along with high precision and recall scores.

The confusion matrix indicates a substantial improvement in identifying churned customers, with far fewer false negatives compared to the baseline model. The recall score for churn prediction reached 0.74, meaning the model successfully identified most customers at risk of leaving.

From a business perspective, this improvement is highly valuable, as accurately detecting at-risk customers enables timely retention strategies and helps reduce potential revenue loss. Overall, the Random Forest model proved to be the most effective approach for churn prediction in this analysis.

6.3. Feature importance analysis

Feature importance analysis from the Random Forest model identified total daytime usage as the most influential predictor of customer churn, followed by customer service calls and international plan status. These findings strongly align with the results from the exploratory data analysis, reinforcing the importance of behavioral and service-related factors in churn prediction.

Additional usage-related variables, such as evening and international call metrics, showed moderate influence, while demographic and account-related variables demonstrated relatively limited predictive power. Overall, the results indicate that customer churn is primarily driven by usage behavior and service experience rather than demographic characteristics.

6.4. Model Performance Evaluation

Model performance was evaluated using accuracy, precision, recall, and F1-score. Particular attention was given to recall, as correctly identifying customers at risk of churn is more critical than minimizing false positives in this business context.

7. Business Recommendations

Based on the findings from exploratory data analysis and predictive modeling, several actionable recommendations can be proposed to reduce customer churn:

7.1. Strengthen Customer Service Strategy

Customers with frequent customer service interactions showed the highest churn risk. Organizations should implement proactive retention strategies for customers with repeated service calls by:

- Improving issue resolution speed
- Monitoring complaint frequency
- Prioritizing high-risk customers for follow-up support

7.2. Target High-Risk Customer Segments

Customers with international plans demonstrated a higher likelihood of churn. Businesses can reduce churn within this segment by:

- Offering customized pricing plans
- Providing loyalty incentives
- Enhancing service quality for international calls

7.3. Focus on High-Usage Customers

Heavy users, particularly those with high daytime usage, were more likely to churn. Companies should:

- Design personalized pricing packages
- Offer usage-based discounts

- Introduce flexible service plans tailored to heavy users

7.4. Promote Value-Added Services

Customers subscribed to additional services such as voice mail plans showed lower churn rates. Businesses should:

- Encourage bundling strategies
- Promote add-on services
- Increase customer engagement through service personalization

8. Conclusion

This study analyzed customer churn behavior using a telecom dataset through exploratory data analysis and machine learning techniques. The results revealed that churn is primarily driven by customer behavior and service-related factors rather than demographic characteristics.

Among all predictors, customer service calls, international plan status, and total daytime usage emerged as the most influential drivers of churn. The Random Forest model demonstrated strong predictive performance and proved effective in identifying customers at risk of leaving.

These findings provide valuable insights for decision-makers and highlight the importance of proactive retention strategies, targeted interventions, and customer-focused service improvements in reducing churn and supporting long-term business growth.

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